Functional Annotation Report: metF (Q88D51, PP_4977) — Methylenetetrahydrofolate Reductase of *Pseudomonas putida* KT2440

Gene/Protein Identity Verification

Before research, the target identity was confirmed against the supplied UniProt record and independent database retrieval:

Attribute	Value
UniProt accession	**Q88D51**
Gene name	**metF** (ordered locus **PP_4977**)
Organism	*Pseudomonas putida* (strain ATCC 47054 / DSM 6125 / KT2440)
Protein	Methylenetetrahydrofolate reductase (MTHFR)
EC number	**1.5.1.54**
Length / mass	295 aa / ~33.0 kDa
Cofactor	**FAD** (one per subunit; ChEBI:57692)
Family	Methylenetetrahydrofolate reductase family
Domains	Pfam **PF02219** (MTHFR); InterPro IPR004620 (MTHF_reductase, bacterial), IPR003171 (Mehydrof_redctse-like), IPR029041 (FAD-linked oxidoreductase-like)
KEGG	ppu:PP_4977

Verification outcome: The gene symbol *metF*, the MTHFR protein description, the FAD flavoprotein family, and the domain architecture are fully mutually consistent. *metF* is the universally used symbol for the bacterial methylenetetrahydrofolate reductase gene (as in *E. coli*, *Salmonella*, *Streptomyces*). This is an unambiguous identification. The literature on the bacterial *metF*/MTHFR family (especially the well-characterized *E. coli* MetF) is directly applicable, and a direct sequence comparison (below) confirms Q88D51 is a bona fide member of this family. Note: the mammalian *MTHFR* gene shares the enzyme name but encodes a larger, differently regulated enzyme (NADPH-specific, with a regulatory domain); it is a distinct ortholog family member and is used here only where the shared catalytic chemistry is relevant.

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1. Summary (Answer to the Research Question)

metF (PP_4977, Q88D51) encodes **methylenetetrahydrofolate reductase (MTHFR; EC 1.5.1.54)**, a **cytoplasmic FAD-dependent flavoenzyme** that catalyzes the **NADH-dependent reduction of 5,10-methylenetetrahydrofolate (CH_2-THF) to 5-methyltetrahydrofolate (CH_3-THF)**. This is the **penultimate, committed step of de novo L-methionine biosynthesis**: the 5-methyl-THF product is the one-carbon (methyl) donor used by methionine synthase (MetH/MetE) to remethylate homocysteine to methionine. The enzyme therefore sits at the branch point of folate one-carbon metabolism that channels one-carbon units into methyl-group (methionine/S-adenosylmethionine) metabolism.

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2. Primary Function: The Catalyzed Reaction and Substrate Specificity

Reaction (physiological direction, reductive):

> 5,10-methylene-THF + NADH + H^+ → 5-methyl-THF + NAD^+ (Rhea:19821; EC 1.5.1.54)

Substrate specificity and cofactors (evidence): - **Folate substrate:** (6R)-5,10-methylene-5,6,7,8-tetrahydrofolate; **product:** (6S)-5-methyl-5,6,7,8-tetrahydrofolate (UniProt catalytic-activity annotation, ChEBI-defined stereochemistry). - **Electron donor: NADH**

(bacterial MetF-type enzymes are NADH-specific, in contrast to NADPH-specific eukaryotic MTHFR). - **Prosthetic group:** a non-covalently but tightly bound **FAD**, one per subunit, which acts as the redox intermediary between NADH and folate (UniProt COFACTOR annotation; keyword "Flavoprotein"). - The closely related, biochemically characterized *E. coli* enzyme "catalyzes the reduction of 5,10-methylenetetrahydrofolate (CH_2-H_4folate) to 5-methyltetrahydrofolate (CH_3-H_4folate)" and is a flavoprotein (Trimmer et al., 2001, P 11371182).

The general family reaction is "the reduction of methylenetetrahydrofolate to methyltetrahydrofolate" (Yu et al., 2025, P 40016375).

Mechanism (inferred from the characterized homolog): Catalysis proceeds by a two–half-reaction (ping-pong) mechanism through the FAD: in the **reductive half-reaction** NADH reduces FAD to $FADH^-$; in the **oxidative half-reaction** the reduced flavin transfers a hydride to CH_2-THF (via a proposed 5-iminium cation intermediate) to give CH_3-THF. In *E. coli*, an active-site aspartate (Asp120) sits near the flavin N1–C2=O locus — a feature that "distinguishes MTHFR from all other known flavin oxidoreductases" — and is required for folate reduction and stabilization of the iminium intermediate; glutamate 28 also contributes to folate activation/catalysis (Trimmer et al., 2001, P 11371182).

In the physiological direction the reaction is essentially irreversible.

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3. Structural Basis and Bioinformatic Evidence for Function

Fold and quaternary structure: The bacterial MTHFR catalytic domain is a **$\beta_8\alpha_8$ (TIM) barrel** that binds one FAD; this fold is conserved from *E. coli* to *Thermus thermophilus*, "quite similar to the β(8)α(8) barrel of E. coli MTHFR," with "one flavin adenine dinucleotide (FAD) prosthetic group bound per dimer" (Igari et al., 2011, P 21858212).

The *E. coli* MetF enzyme assembles as a **homotetramer** (the same study refers to "the tetramer of E. coli MTHFR"; Igari et al., 2011,

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P 21858212

and FAD binding is coupled to large intersubunit conformational changes — so PP_4977 is expected to be an oligomeric (tetrameric) flavoprotein rather than acting as a monomer.

Direct evidence that Q88D51 is a true MetF-type MTHFR (this analysis): - Q88D51 (295 aa) is essentially the **same length as *E. coli* MetF (296 aa)**; it is a **"short," catalytic-domain-only bacterial MTHFR** and lacks the ~300-residue C-terminal regulatory (SAM-binding) domain found in eukaryotic MTHFR. - A Needleman–Wunsch global alignment of Q88D51 vs. *E. coli* MetF (UniProt P0AEZ1) gave **40.7 % identity over 290 aligned residues** — well within the ortholog range for this family. - The two experimentally validated *E. coli* catalytic residues are **strictly conserved** in Q88D51: **Glu28 → Glu24** and **Asp120 → Asp116** (plus flanking conserved positions, e.g., E. coli Pro122 → Pro118, Tyr60 → Tyr56).

Because the complete catalytic apparatus is conserved, the substrate specificity, cofactor usage, and mechanism established for *E. coli* MetF can be transferred to Q88D51 with high confidence.

Regulatory implication of the "short" architecture: Lacking the eukaryotic C-terminal regulatory domain, Q88D51 is **not expected to be allosterically inhibited by S-adenosylmethionine (SAM)**, unlike human MTHFR. Regulation of the bacterial enzyme is exerted primarily at the level of gene expression (Section 5).

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4. Subcellular Localization

The gene product is a **soluble cytoplasmic enzyme**. No signal peptide or transmembrane segment is annotated for Q88D51, and MTHFR-family enzymes are cytosolic flavoproteins. This localization is expected because both substrates — 5,10-methylene-THF (generated by cytoplasmic serine hydroxymethyltransferase/one-carbon metabolism) and NADH — are cytoplasmic, as is the downstream methionine-synthase reaction. The enzyme thus carries out its function in the cytosol, within the folate one-carbon/methionine biosynthetic machinery. (Inference from sequence features and family biochemistry; no experimental localization study specific to PP_4977 was found.)

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5. Pathway Context and Regulation

Pathway membership (UniProt): - *Amino-acid biosynthesis; L-methionine biosynthesis via de novo pathway.* - *One-carbon metabolism; tetrahydrofolate interconversion.*

Position in the pathway: In prokaryotes, "methylenetetrahydrofolate reductase and cobalamin-dependent methionine synthase catalyze the penultimate and ultimate steps in the biosynthesis of methionine" (Matthews et al., 1998, P 9587027).

MetF produces CH_3-THF, which methionine synthase (MetH, B12-dependent; or MetE, B12-independent) uses to convert homocysteine → methionine; methionine is then the precursor of SAM, the cell's principal methyl donor.

Essentiality: Disruption of *metF* causes methionine auxotrophy — "Disruption of the metF gene led to methionine auxotrophy" in *Streptomyces lividans* (Blanco et al., 1998, P 9515933) — demonstrating that the folate branch through MetF is required for methionine synthesis in bacteria. High-throughput RB-TnSeq fitness analysis in *P. putida* KT2440 (Borchert et al., 2024, P 38323821) provides genome-scale functional-module context consistent with such biosynthetic gene assignments.

Pathway reconstruction in *P. putida* KT2440 (this analysis, via KEGG): KEGG assigns PP_4977 to orthology **K00297** (methylenetetrahydrofolate reductase, NADH; EC 1.5.1.54), pathway **ppu00670 "One carbon pool by folate"** (chromosome 5,669,524–5,670,411). The strain encodes a complete methionine/one-carbon route around MetF: serine hydroxymethyltransferase **glyA** (PP_0322, PP_0671) generates the 5,10-CH_2-THF substrate; **both** a B12-dependent methionine synthase **metH** (PP_2375) and B12-independent **metE** (PP_2698, PP_4637) consume the 5-methyl-THF product; and **metK** SAM synthetase (four paralogs) converts the resulting methionine to SAM. Because KT2440 has both MetH and MetE, MetF's 5-methyl-THF output is required for methionine synthesis **regardless of vitamin-B12 availability**.

Note — a second MTHFR paralog: KT2440 carries a second K00297-annotated gene, **PP_4638**, but it is a divergent **495-aa two-domain** protein (only 36.8% identity to *E. coli* MetF and 39.0% to PP_4977) located next to *metE* (PP_4637). PP_4977/Q88D51 — a single-domain ~33 kDa protein with the highest similarity to the characterized *E. coli* MetF — is therefore the **canonical, housekeeping short-form MetF**, while PP_4638 likely represents a specialized/regulated MTHFR-family variant.

Transcriptional regulation (from bacterial paradigms): *metF* is part of the methionine regulon. In enterobacteria the MetR activator coordinates *metE/metH* and modulates the *metF* branch, with vitamin-B12 signaling (MetR "acts in trans to activate the metE and metH genes"; Urbanowski et al., 1987, P 3549685).

In actinomycetes the activator NdgR binds *metH* and *metF* upstream regions ("NdgR was enriched from crude cell lysates with a strong affinity to metH and metF upstream sequences"; Kim et al., 2012, P 23065973).

This places *metF* expression under control tied to sulfur-amino-acid demand rather than to any pleiotropic housekeeping program.

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6. Supported and Refuted Hypotheses

Supported: 1. Q88D51 is a genuine FAD-dependent MTHFR (EC 1.5.1.54) — supported by domain signatures, 40.7 % identity to *E. coli* MetF, and conservation of both catalytic residues. 2. Primary function = NADH-dependent reduction of CH_2-THF to CH_3-THF — supported by UniProt catalytic activity and homolog biochemistry. 3. Physiological role = penultimate step of de novo methionine biosynthesis (essential; auxotrophy on knockout in relatives) — supported. 4. Cytoplasmic, soluble localization — supported by absence of targeting signals and family biochemistry (inference).

Refuted / not applicable: - Q88D51 is **not** a SAM-regulated long-form MTHFR (it lacks the regulatory domain), so allosteric SAM inhibition described for eukaryotic MTHFR does **not** apply. - It is not a membrane, transport, or structural protein.

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7. Limitations and Future Directions

- **No PP_4977-specific biochemical or structural study exists.** Function is assigned by (i) authoritative UniProt/EC annotation, (ii) strong homology and catalytic-residue conservation with the biochemically dissected *E. coli* MetF, and (iii) the conserved bacterial MTHFR fold. Direct kinetic characterization (Km for CH_2-THF and NADH, kcat, cofactor stoichiometry) of the purified *P. putida* enzyme would confirm the transferred parameters.
- **Localization is inferred**, not experimentally measured for PP_4977.
- **Regulation** is described from other bacteria; the specific *cis*/*trans* regulators of *P. putida metF* have not been mapped here.
- A recently described variant subfamily (S6MTHFR; Yu et al., 2025, P 40016375) runs the reaction in the **oxidative** direction coupled to THF-dependent demethylases. Q88D51's high similarity to canonical MetF and its methionine-biosynthesis pathway annotation indicate it is a **conventional reductive MTHFR**, but confirming reaction directionality experimentally would fully exclude the alternative.

References (PMIDs)

- Trimmer, Ballou, Ludwig, Matthews (2001) *Biochemistry* — E. coli MTHFR catalysis; Asp120/Glu28. PMID **11371182**
- Igari et al. (2011) — *T. thermophilus* MTHFR crystal structure; β8α8 barrel, FAD. PMID **21858212**
- Matthews, Sheppard, Goulding (1998) — MTHFR & methionine synthase review. PMID **9587027**
- Blanco, Coque, Martin (1998) — *metF* disruption → methionine auxotrophy. PMID **9515933**
- Yu et al. (2025) — MTHFR family / reverse-reaction subtype. PMID **40016375**
- Kim et al. (2012) — NdgR regulation of *metH/metF*. PMID **23065973**
- Urbanowski et al. (1987) — MetR regulatory gene. PMID **3549685**
- Borchert et al. (2024) — RB-TnSeq functional modules in *P. putida* KT2440. PMID **38323821**
- Primary data: UniProt Q88D51; *E. coli* MetF P0AEZ1 (sequence alignment performed in this study).